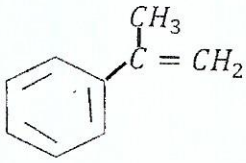
Organic Chemistry Revision (S6 2020)

Mechanisms

1. Complete the following equations and in each case write the accepted mechanism for the reaction.

a) $CH_3CH_2\overset{O}{\overset{\|}{C}} - H \quad + \quad NaHSO_3 \longrightarrow$

b) 

$\xrightarrow{HI}$

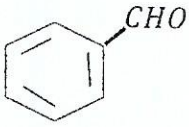
c) $CH_3\underset{Br}{\underset{|}{C}H} - \overset{Br}{\overset{|}{C}}HCH_3 \xrightarrow[\textbf{heat}]{CH_3CH_2\bar{O}Na^+/\,CH_3CH_2OH}$

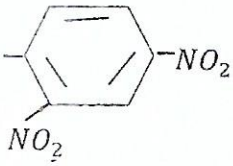
d) $CH_3CH_2 - C \equiv CH \xrightarrow{HBr}$

e) $\xrightarrow[\textbf{Heat}]{conc.\,H_2SO_4/HNO_3}$

f) $CH_3CH_2\overset{Br}{\underset{CH_3}{\underset{|}{\overset{|}{C}}}}CH_3 \xrightarrow[\textbf{Heat}]{NaOH_{(aq)}}$

g) $CH_3CH = CH_2 \xrightarrow{Br_2/H_2O}$

h) 

$H_2N - NH -$ 

$\xrightarrow{H^+}$

i) $+ \quad Br_2 \xrightarrow[\textbf{Heat}]{Fe}$

j) $CH_3CHO \quad + \quad NaHSO_{3(aq)} \longrightarrow$

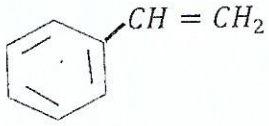
k) $- CH_2I \quad + \quad \bar{O}H_{(aq)} \xrightarrow{Heat}$

l) $+ \quad conc.\,HNO_3 \xrightarrow[\textbf{Heat}]{conc.\,H_2SO_4}$

m) $CH_3CH = CH_2 \xrightarrow{H^+/H_2O}$

n) $CH_3\overset{OH}{\overset{|}{C}}H\,CH_3 \xrightarrow[\textbf{Heat}]{conc.\,H_2SO_4}$

o) $CH_3\overset{O}{\overset{\|}{C}} - H \;+\; \xrightarrow{NaHSO_{3(aq)}}$

p) 

$\xrightarrow{HI}$

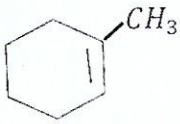
q) $CH_3CH_2CH = CH_2 \xrightarrow{conc.\,H_2SO_4/H_2O}$

r) $CH_3CH_2CH_2COOH \;+\; CH_3CH_2OH \xrightarrow[\text{Heat}]{conc.\,H_2SO_4}$

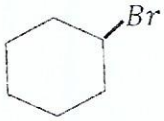
s) $CH_3CH_2\overset{OH}{\overset{|}{C}}HCH_2\,CH_3 \xrightarrow[\textbf{180°C}]{excess\ conc.\,H_2SO_4}$

t) $CH_3CH_2CH_2Br \xrightarrow[\textbf{Heat}]{NaOH_{(aq)}}$

u) $(CH_3)_2NH \;+\; CH_3\overset{O}{\overset{\|}{C}} - Br \longrightarrow$

v) 

$\xrightarrow{HBr}$

w) 

$\xrightarrow{NaOH_{(aq)}}$

x)
$\xrightarrow{\text{Fuming } H_2SO_4}$

y)
$+\; (CH_3)_2C = CH_2 \xrightarrow{H_2SO_{4(aq)}}$

z) $CH_3CH_2Cl \;+\; HC \equiv C^-Na^+ \xrightarrow{liq.\,NH_3/Dry\ ether}$

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